

UNIT I: PHYSICAL OPTICS — INTERFERENCE, DIFFRACTION & POLARIZATION

Weightage: 25% | Syllabus Section 1

1. Wave Interference & Thin Films

Fringe Width (Fresnel Biprism): $\beta = \frac{\lambda D}{2d}$, where $2d$ is virtual slit separation.

Thin Film Interference: Optical path diff: $\Delta =$

$2\mu t \cos r \pm \frac{\lambda}{2}$.

• Maxima (Constructive): $2\mu t \cos r = (2n - 1)\frac{\lambda}{2}$

• Minima (Destructive): $2\mu t \cos r = n\lambda$

Newton's Rings: Ring diameter: $D_n^2 = 4nR\lambda$ (Dark ring),

$D_n^2 = 2(2n - 1)R\lambda$ (Bright ring). Wavelength: $\lambda =$

$\frac{D_{n+p}^2 - D_n^2}{4pR}$.

2. Fraunhofer Diffraction & Grating

Single Slit Minima: $a \sin \theta = n\lambda$ ($n = \pm 1, \pm 2, \dots$).

Central Max width: $2\lambda D/a$.

Plane Transmission Grating: Principal Maxima condition: $(a + b) \sin \theta = n\lambda$.

• Grating Element: $a + b = \frac{1}{N}$ (N lines per cm).

• Absent Spectra condition: $\frac{a+b}{a} = \frac{n}{m}$.

Resolving Power of Grating: $\frac{\lambda}{\Delta\lambda} = n \cdot N$, where N is total ruling lines.

3. Polarization & Optical Activity

Brewster's Law: Polarizing angle i_p : $\mu = \tan i_p$.

Reflected and refracted rays are mutually perpendicular: $i_p + r = 90^\circ$.

Malus' Law: $I = I_0 \cos^2 \theta$, where θ is angle between polarizer and analyzer.

Retardation Plates:

• Quarter-Wave Plate (QWP): $t = \frac{\lambda}{4(\mu_e - \mu_o)}$, phase diff $\pi/2$

• Half-Wave Plate (HWP): $t = \frac{\lambda}{2(\mu_e - \mu_o)}$, phase diff π .

Specific Rotation: $[\alpha]_\lambda^T = \frac{\theta}{LC}$ (L in dm, C in g/cm³).

UNIT II: QUANTUM MECHANICS & SCHRÖDINGER EQUATION

Weightage: 25% | Syllabus Section 2

1. De Broglie Waves & Uncertainty Principle

Matter Wave Relations: $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2mqV}}$.

Electron Wavelength: $\lambda = \frac{1.227}{\sqrt{V}}$ nm (V in Volts).

Heisenberg Uncertainty: $\Delta x \cdot \Delta p_x \geq \frac{h}{2}$; $\Delta E \cdot \Delta t \geq \frac{h}{2}$; $\Delta \theta \cdot \Delta J \geq \frac{h}{2}$.

Application: Proves non-existence of free electrons in atomic nucleus!

2. Schrödinger Wave Equations

Time-Dependent (1D): $i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x)\Psi$.

Time-Independent (Steady State): $\frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0$.

Born's Probabilistic Interpretation: Probability density $P(x) = |\psi(x)|^2 = \psi^* \psi$.

• Normalization condition: $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$.

3. Particle in a 1D Potential Box

Boundary Conditions: $V(x) = 0$ for $0 < x < L$; $V(x) = \infty$ outside.

Normalized Wavefunction: $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$ ($n = 1, 2, 3, \dots$).

Quantized Energy Levels: $E_n = \frac{n^2 h^2}{8mL^2} = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$.

• Zero-Point Ground Energy: $E_1 = \frac{h^2}{8mL^2} \neq 0$.

UNIT III: LASERS, FIBER OPTICS & SUPERCONDUCTIVITY

Weightage: 25% | Syllabus Section 3

1. Laser Physics & Einstein Coefficients

Einstein Relations: $B_{12} = B_{21}$ (Absorption = Stimulated Emission probability). Ratio: $\frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3}$.

Three Crucial Conditions: (1) Population Inversion ($N_2 > N_1$), (2) Metastable state (lifetime 10^{-3} s), (3) Optical Resonant Cavity ($L = n\lambda/2$).

Ruby Laser: 3-level (Cr^{3+} ions in Al_2O_3 , 694.3 nm red pulsed).

He-Ne Laser: 4-level gas laser (He:Ne = 10:1, 632.8 nm continuous red).

2. Optical Fiber Communication

Total Internal Reflection (TIR): Angle of incidence $\theta_i > \theta_c$, where critical angle $\sin \theta_c = n_2/n_1$ ($n_1 > n_2$).

Numerical Aperture (NA): Light gathering power: $NA = \sin \theta_a = \sqrt{n_1^2 - n_2^2} \approx n_1 \sqrt{2\Delta}$, where $\Delta = \frac{n_1 - n_2}{n_1}$.

V-Number / Normalized Frequency: $V = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi a}{\lambda} NA$.

• Single-Mode condition: $V < 2.405$. Number of modes in MMF: $M \approx \frac{V^2}{2}$.

3. Superconductivity & Meissner Effect

Critical Parameters: Resistance drops to zero below critical temp T_c . Critical magnetic field: $H_c(T) = H_c(0) \left[1 - \left(\frac{T}{T_c}\right)^2\right]$.

Meissner Effect: Perfect diamagnetism inside superconductor: $B = \mu_0(H + M) = 0 \implies \chi = \frac{M}{H} = -1$.

Type I vs Type II:

- Type I: Complete Meissner, single H_c , low field applications.
- Type II: Two critical fields (H_{c1} , H_{c2}), mixed vortex state between them, high-field superconducting magnets.

UNIT IV: DIELECTRIC MATERIALS, MAGNETIC PROPERTIES & MAXWELL EQUATIONS

Weightage: 25% | Syllabus Section 4

1. Dielectrics & Polarization Vectors

Polarization Vector P: $\vec{P} = \epsilon_0 \chi_e \vec{E}$. Electric

Displacement: $\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_r \epsilon_0 \vec{E}$.

Claussius-Mossotti Equation: Connects microscopic polarizability α to macroscopic ϵ_r : $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$.

Polarization Mechanisms: (1) Electronic (α_e , temp-independent), (2) Ionic (α_i), (3) Orientational ($\alpha_o = \frac{p^2}{3kT}$, strongly temp-dependent).

2. Magnetic Classification & Hysteresis

Magnetic Vectors: $\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_r \mu_0 \vec{H}$;

Susceptibility $\chi_m = \frac{M}{H} = \mu_r - 1$.

Diamagnetic: $\chi_m < 0$, $\mu_r < 1$, temp independent.

Paramagnetic: $\chi_m > 0$, follows Curie Law $\chi_m = \frac{C}{T}$.

Ferromagnetic: Spontaneous domains, Curie-Weiss law $\chi_m = \frac{C}{T - \theta}$.

B-H Hysteresis Loop: Retentivity (B_r , at $H = 0$),

Coercivity (H_c to demagnetize). Area of loop = Hysteresis energy loss per cycle!

3. Maxwell's Electromagnetic Equations

Law	Differential Form
Gauss Law (E)	$\nabla \cdot \vec{D} = \rho_v$
Gauss Law (B)	$\nabla \cdot \vec{B} = 0$ (No monopoles)
Faraday's Law	$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
Ampere-Maxwell	$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

Displacement Current: $J_d = \frac{\partial D}{\partial t} = \epsilon \frac{\partial E}{\partial t}$. EM Wave speed: $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8$ m/s.